GATE

Computer Science & Information Technology

Computer Networks

DPP: 2

ISO-OSI Stack

Q1 A packet has arrived with an MF(More fragments) bit value of 1. Is this the first fragment, the last fragment, or a middle fragment?
(A) First fragment
(B) Middle fragment
(C) Last fragment
(D) More information required

Q2 In an IPV4 packet, the value of the HLEN field is 1100 in binary. How many bytes of options field is being carried by this packet?
(A) 28 byte (B) 20 byte
(C) 8 byte (D) 40 byte

Q3 What is the maximum payload size at the Network layer and the Data link layer on the TCP/IP protocol?
(A) Any size, 1480 bytes
(B) 65,495 bytes, 1500 bytes
(C) 65,535 bytes, 1500 bytes
(D) 65,515 bytes, 1500 bytes

Q4 Consider an IP link with a maximum transfer unit of 700 bytes, and each datagram size to send is 2400 bytes. Assume that the datagram recently sent has an identification number of 422. What is the total number of fragments that must be sent to transfer the full datagram, and what is the offset value of the 2nd fragment from all the fragmented datagrams? Note: Consider the minimum header size of the IPV4 datagram.
(A) 4, 85 (B) 5, 680
(C) 4, 1320 (D) 1,0

Q5 Which of the following assertions is/are TRUE about the Internet Protocol (IP)?
(A) It is possible for a computer to have multiple IP addresses.
(B) IP packets from the same source to the same destination can take different routes in the network.
(C) IP ensures that a packet is discarded if it is unable to reach its destination within a given within a given umber of hops.
(D) The packet source cannot set the route of an outgoing packets; the route is determined only by the routing tables in the routers on the way.

Q6 Suppose a packet of 1800 bytes with minimum header size enters a local area network with an MTU (Maximum transfer unit) of 500 bytes. Then, how many fragments will be generated from the packet, and what is the payload size of the last packet?
(A) 3, 300 (B) 4, 340
(C) 5, 320 (D) 3, 340

Q7 If the value in the HLEN field of the IP header is 8, then what is the header's length?
(A) 8 bytes (B) 16 bytes
(C) 32 bytes (D) 40 bytes

Q8 A TCP message consisting of 2800 bytes is passed to IP for delivery across two networks. The first network can carry a maximum payload of 1500 bytes per frame, and the second network can carry a maximum payload of 500

Android App | iOS App | PW Website

GATE

bytes per frame, excluding network overhead. Assume that the IP overhead per packet is 20 bytes. What is the total IP overhead in the second network for this transmission?

(A) 140 bytes
(B) 80 bytes
(C) 120 bytes
(D) 160 bytes

Android App | iOS App | PW Website

Answer Key

Q1	(D)	Q5	(A, B, C)
Q2	(A)	Q6	(B)
Q3	(D)	Q7	(C)
Q4	(A)	Q8	(A)

Android App | iOS App | PW Website

Hints & Solutions

Q1 Text Solution:

If the M bit is 1, it means that there is at least one more fragment. one or the middle one, but not the last one. We don't know if it is the first or the middle one; we need more information (the value of the fragmentation offset). However, the original packet has been fragmented because the M bit value is 1.

MF = 0 Offset = 0 No fragment

MF = 0 Offset = 10 → Last fragment

MF = 1 Offset 0 →First fragment

MF = 1 Offset 10 → Middle fragment

This fragment can be the first

Q2 Text Solution:

The HLEN value is 12 in decimal, which means the length of the header is 12 × 4 = 48 bytes.

The minimum header length is 20 bytes, and the remaining will be of options filed.

Therefore, 48 – 20 = 28 bytes are the options field.

Q3 Text Solution:

Payload means only data without the header.

The application layer can forward any data size to the transport layer.

The transport layer can hold the maximum size of the data (payload) is 65,495 bytes.

The Network layer can hold the maximum size of the data, which is 65,515 bytes.

The data link layer can hold the maximum size of the data is 1500 bytes.

Q4 Text Solution:

The Maximum transfer Unit = 700 bytes

Each datagram size = 2400 bytes

Minimum size of IPV4 header = 20 bytes

2400 bytes = 2380 bytes data + 20 bytes header

Then the datagram divides into multiple frames depending on MTU

680 bytes data + 20 bytes header → fragment 1

680 bytes data + 20 bytes header → fragment 2

680 bytes data + 20 bytes header → fragment 3

370 bytes data + 20 bytes header → fragment 4

→ Total number of fragments = 4

Offset is the number of data bytes ahead of a particular fragment.

Offset of 2nd fragment is 680/8 = 85

Q5 Text Solution:

A is true since a computer can have multiple IP addresses for each interface.

B is true since IP is a connectionless protocol that uses the datagram approach, which means that each datagram is handled independently and can follow a different route to the destination.

C is true since if there is ttl associated with each packet, and it means the maximum no of hops it can have. When it becomes zero, the packet is discarded by the intermediate router, and time exceeded message is sent to the source to inform it about the situation.

D is false since a source can predetermine a route for the datagram as it travels through the internet. This is done with the help of a strict source routing option.

Q6 Text Solution:

Total data bytes = 1800-20-1780

Data bytes in first fragment = 500 -20 = 480

Data bytes in second fragment = 500 -20 = 480

Data bytes in third fragment = 500 -20 = 480

Remaining data bytes = 340

Hence size of the last packet = 340 + 20 = 360

Android App | iOS App | PW Website

It asks for payload size, and in the payload, only data is contained, not the header. i.e. 340

Q7 Text Solution:

Length of header is 8 x 4 = 32 bytes

Q8 Text Solution:

In the question, the maximum payload that can be transferred is given, not the maximum packet size. So, during fragmentation, we must consider only the payload, not the headers. At first network, 2800 byte payload will be divided into 1500 byte and 1300 bytes payloads

Because we can not fragment sizes 1500 and 1300 because these are not divisible by 8.

Fragmentation at the 1st network, 1496 and 1304

Fragmentation at the 2nd network, 1496 will be divided into 4 packets of 496,496,496 and 8 (80 bytes overhead), and 1304 will be divided into three packets of 496,496 and 312 (this gives 60 bytes overhead)

Therefore, a total of 7 fragmentations. so total IP overhead = 7 x 20=140 bytes

Android App | iOS App | PW Website